



L-SYSTEM INTERVALS – THE PICTURE, AND THE PROGRAM THAT MADE IT

DOC NO. RL-Z0-LSYSTEMS-A REV. A RIPOSTE LABORATORIES

A second family of station intervals, alongside the Glitchsheet ones in `intervals.md`. Each clip is a Lindenmayer system: a handful of rewriting rules, expanded, read back as turtle orders, and drawn – with its own source code set beside it on screen.

CONTENTS

▶ One file, three jobs	3
▶ What is on screen	4
▶ The specimens	5
▶ The pipeline – one host, unlike the Glitchsheet intervals	6
▶ Why the picture moves at all	7
▶ Traps	8
▶ Getting them on air	9
▶ Tuning	10

The two halves of the frame cannot disagree, because they are the same file.

SEC.01

ONE FILE, THREE JOBS

Every specimen in `tools/z0-lsys/specimens/` is a self-contained JavaScript file, used three times over:

USED BY	AS	WHAT IT DOES
<code>qjs lsys-emit.js</code>	an ES module	walks the turtle; the renderer draws those strokes as the ink
<code>ffedit -f mv -s <file></code>	an FFglitch script	replaces every motion vector with the heading of the branch crossing that macroblock
the renderer	text	prints it, verbatim, into the panel on the right

So the code on screen is not an illustration of the algorithm. It is the algorithm – the same bytes that `ffedit` ran, with the `sha1` shown in the footer alongside the exact command and its `-sp` payload.

Nothing is hidden behind an `import`. That is the whole claim, and it is why the machinery is duplicated into all eight files rather than shared. The files are therefore **output**: edit `specimen.template.js` or the `SYSTEMS` table in `make-specimens.py` and regenerate. Hand-edits to `specimens/*.js` are lost.

SH

`tools/z0-lsys/make-specimens.py`

The generator enforces a **52-column limit** and refuses to write anything if a single line is over it. That is not style: the panel is 406 px wide and the listing is set at 12 px, so a longer line is a line nobody can read. It checks every file before writing any of them – validating on the way out once left the over-wide file on disk next to the complaint about it, and the render used it regardless.

SEC.02

WHAT IS ON SCREEN

```

+-----+-----+
|          | CH 0  STATION INTERVAL |
|  448 x 480  | FERN                      |
|  the L-system, drawn  | six rewrites of one  |
|  stroke by stroke and  | -----|
|  smeared by its own  | export const SYS = {  |
|  motion vectors      |   axiom: "X",         |
|                      |   rules: { X: "F+[[X.. |
|                      |   ...                 |
|                      | -----|
|                      | F+[[X]-X]-F[-FX]+X ... | <- the live string
|                      | $ ffedit -f mv -s ... |
|                      | gen 6  25,159 symbols |
+-----+-----+

```

854 x 480, the channel's own frame – nothing is pillarboxed

The clip has an arc, because an interval is cut at an arbitrary boundary and has to be worth joining at any point:

SHARE OF CLIP	WHAT HAPPENS
0-3%	the specimen card: name, rules, an empty frame
3-85%	the turtle draws, and a wave of motion travels the branches in the order they were drawn
85-100%	the drawing is finished and drifts, smearing along its own tangents

The source pages through itself **exactly once per clip**, however long the file is: the renderer takes the fewest pages that hold it and then spreads the lines evenly, so the last page is not two lines and eight seconds of empty panel.

The strip under the listing is the real expanded L-string, windowed at the turtle's current position – 147 lines of program above, and the 25,159 symbols it produced running past below.

SEC.03

THE SPECIMENS

SLUG	NAME	RULES	ANGLE	GENS	PUSH	PHOSPHOR
fern	FERN	$X \rightarrow F+[[X]-X]-F[-FX]+X$, $F \rightarrow FF$	25°	6	30	amber
dragon	DRAGON	$X \rightarrow X+YF+$, $Y \rightarrow -FX-Y$	90°	12	15	green
hilbert	HILBERT	$A \rightarrow +BF-AFA-FB+$, $B \rightarrow -AF+BFB+FA-$	90°	6	10	bone
koch	KOCH ISLAND	$F \rightarrow F+F-F-FF+F+F-F$	90°	3	16	red
sierpinski	SIERPINSKI	$F \rightarrow F-G+F+G-F$, $G \rightarrow GG$	120°	6	30	amber
levy	LEVY C	$F \rightarrow +F-F+$	45°	14	15	green
bush	BUSH	$F \rightarrow FF-[-F+F+F]+[+F-F-F]$	22.5°	5	28	amber
crystal	CRYSTAL	$F \rightarrow FF+F+F+F+FF$	90°	4	22	bone

push is per specimen and lives in SYS, on screen, because it has to be: **a space-filling curve fills every macroblock, so the shove that animates a fern turns a Hilbert curve to mud.** Hilbert takes a third of the fern's.

The palette is the phosphor set from BRAND.md_{RL-Z0-A} — amber #fffb0000, phosphor red #ff2d2d, green #33ff66, bone #cfcabc on #0d0d0d. An interval represents the broadcast, so it gets the CRT treatment, not the Riposte brand.

SEC.04 THE PIPELINE – ONE HOST, UNLIKE THE GLITCHSHEET INTERVALS

```
LXC 111 (claude)
qjs      -> the turtle, as geometry
PIL      -> the ink, frame by frame, over a pipe
ffgac    -> MPEG-2 carrier          +nopimb +forcemv, one GOP
ffedit   -> the same specimen file, over the motion vectors
ffgac    -> decode
PIL      -> the source panel beside it
ffmpeg   -> H.264 854x480, silent stereo AAC
```

SH

```
pct exec 111 -- su - user -c \
'cd ~/channel-z0/tools/z0-lsys && python3 z0-lsys-render.py --out-dir /tmp/z0lsys/out'
```

Eight 55-second clips in about **40 seconds**, all told.

`tools/z0-generative.py` needs three hosts because Glitchsheet's engine and numpy live in LXC 114, which has no ffmpeg at all. Nothing here needs either: LXC 111 has `ffgac`, `ffedit` and `qjs` under `/opt/ffglitch` and a real ffmpeg with `libx264`. (numpy in LXC 111 is in fact broken – built against a newer glibc than the container has – so PIL does all the raster work.)

SEC.05

WHY THE PICTURE MOVES AT ALL

The carrier is a **still** as far as the encoder is concerned. New strokes appear, so those macroblocks get a residual; everything already drawn is predicted from the previous frame with a zero residual. A block with no residual is **whatever its motion vector fetched** – and the motion vectors are the L-system. So the decoder becomes the paint engine.

That only works because of the encode flags:

FLAG	WHY
+forcemv	writes a vector for every macroblock even where the encoder wants none
+nopimb	keeps intra blocks out of P frames, so nothing scrubs itself clean
-qscale:v 1, -g <all>, -sc_threshold max	one I frame at the top; damage propagates for the whole clip
448x480	28x30 whole macroblocks – a partial block column at the edge breaks the grid, and the grid is the aesthetic

SEC.06

TRAPS

- ▶ **A motion vector that fetches from outside the frame decodes as flat saturated green.** The specimens fade their field out over the last three macroblocks at every edge for that reason alone. It is in the code on screen, and it is not a taste decision.
- ▶ **-sp takes integers only.** ffedit's JSON parser rejects a float outright and then writes no output file at all, which reads exactly like a crash. Fractions travel percent-scaled and are divided inside the script.
- ▶ **Comparing the two bitstreams proves nothing.** ffedit re-codes a vector it was handed even when the value is unchanged, so a script that sets every vector to zero still writes a **different file** while the picture stands perfectly still. The render guards on the real thing instead: it decodes the untouched carrier alongside the glitched one and refuses a clip whose pictures differ by less than `-min-motion`. Same encoder, same quantiser, so with the vectors left alone the measure is exactly zero. `-trim 0` is the negative test, and it is refused.
- ▶ **Watching the glitched clip on its own says little more,** because the drawing is moving anyway – the turtle is still drawing it.
- ▶ Every clip is checked for length within a second of 55.0 s before it is installed, in the spirit of `tools/z0-weather.sh`.

SEC.07

GETTING THEM ON AIR

The clips carry the `generative` tag in their `.nfo`, so they join the existing interval pool and **need no schedule change**. Drop them in and let the library refresh:

SH

```
| scp z0-lsys-*.mp4 z0-lsys-*.nfo root@10.0.1.222:/mnt/main-data/channelz0/generative/
```

The NFO's tags replace the folder's tags, so `<tag>generative</tag>` has to be in the file – that is what the renderer writes, alongside `media`.

The library refreshes on `scanner.library_refresh_interval` (6 hours), so they join the shuffle on their own. To have them sooner, scan the library from the ErsatzTV UI. A new tag would need more – the pool is a `search:` key answered by Lucene, and an unindexed tag silently yields an empty pool – but `generative` already exists and is already indexed.

SEC.08

TUNING

WANT	CHANGE
A different system	SYSTEMS in make-specimens.py, then regenerate
Calmer or wilder	that specimen's push (it is per specimen for a reason)
Everything calmer	-trim, a global percentage over every push
More or less drift at the end	-bloom
A narrower travelling wave	-win, in % of the string
Longer clips	-seconds; the source still pages through exactly once

SOURCE docs/lssystems.md SHA 054faf89d182 BUILT 2026-08-19